Proposed experiments for searching the stimulation parameter space

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The purpose of these experiments is to investigate how various parameters of charge-balanced (biphasic) stimuli (Fig. 1), including

1. pulse amplitude A,
2. pulse frequency f_{pulse},
3. burst frequency f_{burst}, and
4. duration of the stimulus D

affect the physiology of the brain.

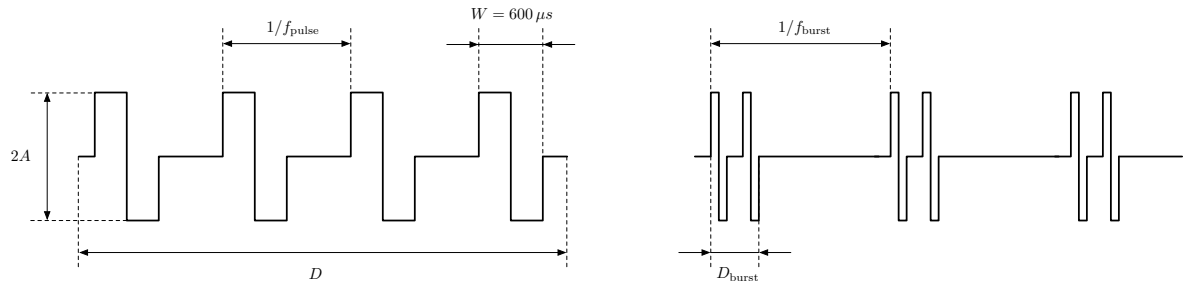


Figure 1: Parameters of the proposed continuous and theta-burst stimuli.

We propose to explore the space of stimulation parameters along several planes described in Sections 2-4 below.

1 Safe Amplitude Determination

Prior to the start of each experiment, we will determine a maximum current amplitude that does not cause after discharges (ADs). Specifically, for each frequency f^i_{pulse} in the set

$$f_{\text{pulse}} = \{f^1_{\text{pulse}}, f^2_{\text{pulse}}, ..., f^N_{\text{pulse}}\}\,\text{Hz},$$

we will sweep the current amplitude from $0 - 1.5\,\text{mA}$ in steps of $0.5\,\text{mA}$ to find such maximum current amplitude A^i_{max}.

For each frequency-current combination, we will deliver a 1.5-second-long pulse with these parameters and wait for the neurophysiologist to check and report whether or not any ADs occurred. If ADs are observed for current amplitude A, the neurophysiologist will be asked to lower the current amplitude by $0.25\,\text{mA}$. If no ADs are reported at this lower value, $A - 0.25\,\text{mA}$ current amplitude will be recorded as the maximum current amplitude. If ADs <u>are</u> observed, then $A - 0.5\,\text{mA}$ will be recorded instead.

If for some frequency f^j_{pulse} ADs are observed for the lowest value of the current, then that frequency value will not be tested in any of the subsequent experiments.

As a secondary check, once a maximum current amplitude A^i_{max} is established for the frequency f^i_{pulse}, we will deliver 5 trains of $1.5\,\text{s}$ stimulation with a $2.75 - 3.25\,\text{s}$ Inter-Stimulus Interval (ISI) at that amplitude and frequency to be certain that repeated stimulation with these parameters is safe.

For $N = 5$ different frequencies and a waiting time of roughly 10 seconds to check for ADs, the entire protocol for finding a safe value of the current should take about $5 \times 4 \times (1.5 + 10) + 5 \times 5 \times (1.5 + 3) \approx 6$ minutes.

The resultant set

$$A_{\text{max}} = \{A^1_{\text{max}}, A^2_{\text{max}}, ..., A^N_{\text{max}}\}\,\text{mA}$$

of current amplitudes will be used in subsequent experiments.

NOTE: Due to hardware limitations of System 1.0, the value of the current amplitude cannot be automatically changed during an experiment. Therefore, we will select a minimum "safe" value

$$A_{\text{safe}} = \min(A_{\text{max}})$$

that does not cause after-discharges across all tested frequencies. Unless otherwise specified, this value of current amplitude will be used in all of the following protocols.

2 Experiment 1: (f_{pulse}, D) plane

In our initial set of experiments, the pulse frequency set will comprise $N = 5$ frequencies (two below 50 Hz and two above 50 Hz):

$f_{\text{pulse}} = \{10, 25, \mathbf{50}, 100, 200\}$ Hz.

The stimulus duration values and the current amplitude will be set to

$D = \{0.25, 0.5, 1\}$ s,

$A = A_{\text{safe}}$.

A stimulus described by a particular (f_{pulse}, D) combination will be separated from the subsequent stimulus by a uniformly distributed Inter-Stimulation Interval (ISI) of 2.75 - 3.25 seconds. The experiment will proceed by delivering the entire set of $5 \times 3 = 15$ different stimuli, which should take roughly $5 \times (3.25 + 3.5 + 4) = 53.75$ s to complete. After every 3 repetitions of the entire set of 15 (f_{pulse}, D) combinations, or roughly every 3 minutes, the neurophysiologist will be prompted to check for ADs. If no ADs are reported, another 3 repetitions of stimuli will be delivered. We would get 30 set repetitions in about 27 minutes.

The neurophysiologist will be able to stop or pause the experiment at any point.

3 Experiment 2: (f_{pulse}, A) plane

As in the above experiment,

$f_{\text{pulse}} = \{10, 25, \mathbf{50}, 100, 200\}$ Hz.

However, we will fix the stimulus duration to

$D = 0.5$ s

and will vary the amplitude A instead. Specifically, we will employ the following set of current amplitudes:

$A = \{A_{\text{safe}} - 0.5\,\text{mA},\ A_{\text{safe}} - 0.25\,\text{mA},\ A_{\text{safe}}\}$.

In addition to the above combinations of frequencies and amplitudes, we will also deliver a stimulus consisting of a single biphasic pulse at each of the above amplitude values.

A stimulus described by a particular (f_{pulse}, A) combination will be separated from the subsequent stimulus by a uniformly distributed Inter-Stimulation Interval (ISI) of 2.75 - 3.25 seconds. The experiment will proceed by delivering the entire set of $6 \times 3 = 18$ different stimuli (including single-pulse stimuli), which should take roughly $5 \times 3 \times 3.5 + 3 \times 3 = 61.5\,\text{s}$ to complete. After every 3 repetitions of the entire set of 18 stimuli, or roughly every 3 minutes, the neurophysiologist will be prompted to check for ADs. If no ADs are reported, another 3 repetitions of all stimuli will be delivered. We would get 30 set repetitions in about 30 minutes.

The neurophysiologist will be able to stop or pause the experiment at any point.

4 Experiment 3: $(f_{\text{pulse}}, f_{\text{burst}})$ plane

In this experiment, we will use

$f_{\text{pulse}} = \{50, 100\}\,\text{Hz}.$

The amplitude will be fixed to

$A = A_{\text{safe}},$

while the duration of each stimulus D will vary depending on the burst frequency f_{burst}. The set of burst frequencies will be

$f_{\text{burst}} = \{3, 4, 5, 6, 7, 8\}\,\text{Hz}.$

The burst duration D_{burst} will be fixed to

$D_{\text{burst}} = 0.25/f_{\text{burst}},$

or 25% of the corresponding burst cycle. A total of 5 bursts will be delivered for each $(f_{\text{pulse}}, f_{\text{burst}})$ combination, resulting in a stimulus duration D that will vary from $5/8 = 0.625\,\text{s}$ for $f_{\text{burst}} = 8\,\text{Hz}$ to $5/3 = 1.667\,\text{s}$ for $f_{\text{burst}} = 3\,\text{Hz}$.

A stimulus described by a particular $(f_{\text{pulse}}, f_{\text{burst}})$ combination will be separated from the subsequent stimulus by a uniformly distributed Inter-Stimulation Interval (ISI) of 2.75 - 3.25 seconds. The experiment will proceed by delivering the entire set of $2 \times 6 = 12$ $(f_{\text{pulse}}, f_{\text{burst}})$ combinations, which should take roughly a minute and a half ($2 \times (5/8 + 5/7 + 5/6 + 5/5 + 5/4 + 5/3 + 6 \times 4) \approx 48.2\,\text{s}$) to complete. After every 10 repetitions of the entire set of 12 $(f_{\text{pulse}}, f_{\text{burst}})$ combinations, or roughly every 8 minutes, the neurophysiologist will be prompted to check for ADs. If no ADs are reported, another

10 repetitions of all waveforms will be delivered. We would get approximately 30 set repetitions in under 25 minutes.

The neurophysiologist will be able to stop or pause the experiment at any point.

5 Summary of Experiments Across Sites for Year 1

We expect that the effects of stimulation will vary on a patient by patient basis (e.g., due to electrode placement, physiological differences, etc.) and would like to perform all three experiments in the same subject whenever possible. This could be accomplished in a course of one or two days.

With experiments running roughy from March-June, we would like to enroll one subject a month, for a total of three subjects at each participating location.

Subjects (3) * Sites (8) = 24 total sessions.

Breakdown across brain regions:
– 40% Hippocampus (60% CA1, 10% CA2/CA3, 30% Dentate Gyrus)
– 25% Entorhinal cortex
– 35% other MTL structures